

Submission STAT1-PS2-SUB02

Question STAT1-PS2-Q01

work / setup:

1. $\bar{x} = 33/5 = 6.60$

crossed-out arithmetic: [smudged]

FINAL (maybe): mean=6.60; sample variance=16.30 plus 1

Question STAT1-PS2-Q02

work / setup:

1. $P(D_1 \cap D_2) = (5/13)((5-1)/(13-1))$

crossed-out arithmetic: [smudged]

FINAL (maybe): 0.1282 plus 1

Question STAT1-PS2-Q03

work / setup:

1. $SE = \sqrt{(p(1-p))/n} = \sqrt{(0.55(1-0.55)/40)} = 0.0787$

2. $CI = 0.55 \pm 1.96(0.0787) = (0.396, 0.704)$

crossed-out arithmetic: [smudged]

FINAL (maybe): (0.396, 0.704); repeated-sampling interpretation plus 1

Question STAT1-PS2-Q04

work / setup:

1. Increase sample size n to reduce standard error.

2. Reduce confidence level or population variability when justified.

crossed-out arithmetic: [smudged]

FINAL (maybe): increase n ; lower confidence level or variability; note the coverage trade-off plus 1

Question STAT1-PS2-Q05

work / setup:

1. $0.03 < 0.05$, so reject H_0 under the stated rule.

crossed-out arithmetic: [smudged]

FINAL (maybe): reject H_0 ; evidence against H_0 , not proof that H_0 is false plus 1

Question STAT1-PS2-Q06

work / setup:

1. $\bar{x} = 29/5 = 5.80$

crossed-out arithmetic: [smudged]

FINAL (maybe): mean=5.80; sample variance=16.70 plus 1

Question STAT1-PS2-Q07

work / setup:

1. $P(D_1 \cap D_2) = (5/19)((5-1)/(19-1))$

crossed-out arithmetic: [smudged]

FINAL (maybe): 0.0585 plus 1

Question STAT1-PS2-Q08

work / setup:

1. $SE = \sqrt{(p(1-p))/n} = \sqrt{(0.45(1-0.45)/80)} = 0.0556$

2. $CI = 0.45 \pm 1.96(0.0556) = (0.341, 0.559)$

crossed-out arithmetic: [smudged]

FINAL (maybe): (0.341, 0.559); repeated-sampling interpretation plus 1